

Feature Selection

Tuesday, 15 September 2026 4:26 PM



AML_Lectur
e10_Work...

Advanced Machine Learning

Worksheet 10 — Lecture 10

ADVANCED MACHINE LEARNING · WORKSHEET 10

Feature Selection

Duplicate Removal · Variance & Correlation Filters · Wrapper Search · Validation Discipline
Lecture 10

Name _____	Branch / Year _____	Date _____
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PART I: WHY FEATURE SELECTION?

SECTION 1 More Columns Do Not Always Mean More Information

► HOOK

In Lecture 8, we deliberately created features such as x^2 so that a model could represent a curved pattern. In Lecture 9, we saw that extra flexibility can reduce bias but can also increase variance. We now ask the complementary question:

When a dataset already contains many columns, which columns deserve to remain?

Return to the internship-stipend problem. The target is a student's stipend, and the available candidate inputs are:

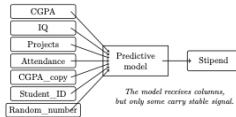
Feature	Why it may appear useful	Question to ask
CGPA	Academic performance may influence shortlisting.	Does it add stable predictive signal?
IQ score	May relate to problem-solving ability.	Does it help after other features are present?
Number of projects	May reflect practical experience.	Is its contribution linear or interaction-based?
Attendance	Could reflect discipline.	Is the relationship real or accidental in this sample?
CGPA_copy	Exactly repeats CGPA.	Does it contribute any new information?
Student_ID	Has large numerical variation.	Is it a genuine predictor or only an identifier?
Random_number	Changes from student to student.	Does variation alone make it relevant?

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Visual: Seven candidate columns enter one prediction problem



Classroom question: Would the model necessarily become better if we keep all seven columns?

SECTION 2 Feature Selection: The Problem and the Mathematical View

A feature is an _____ used by a model to predict a target. Feature selection is the process of choosing a _____ while preserving, or ideally improving, performance on unseen data.

★ KEY INSIGHT

"Most important" is not an absolute label. It depends on the target, model, evaluation metric, sample, and the other features already present.

2.1 A compact mathematical view

Suppose the design matrix contains n rows and p candidate features:

$$X = [f_1 \ f_2 \ \dots \ f_p] \in \mathbb{R}^{n \times p}.$$

Let $S \subseteq \{1, 2, \dots, p\}$ be a subset of feature indices. The notation X_S means that we keep only the columns whose indices are in S .

$$S^* = \arg \max_g \text{CVScore}(\text{Model trained on } X_{S_g} | y)$$

For a loss such as RMSE, use _____ instead of $\arg \max$.

If two subsets have nearly the same cross-validated performance, prefer the _____ unless domain knowledge gives a strong reason not to.

2.2 Three kinds of feature behaviour

Type	Meaning	Running example
Relevant	Carries stable information that helps predict the target.	Number of projects may help predict stipend.
Redundant	Repeats information already carried by another feature.	CGPA_copy repeats CGPA.
Irrelevant	Varies but does not carry a stable relationship with the target.	Random_number may vary and still be useless.

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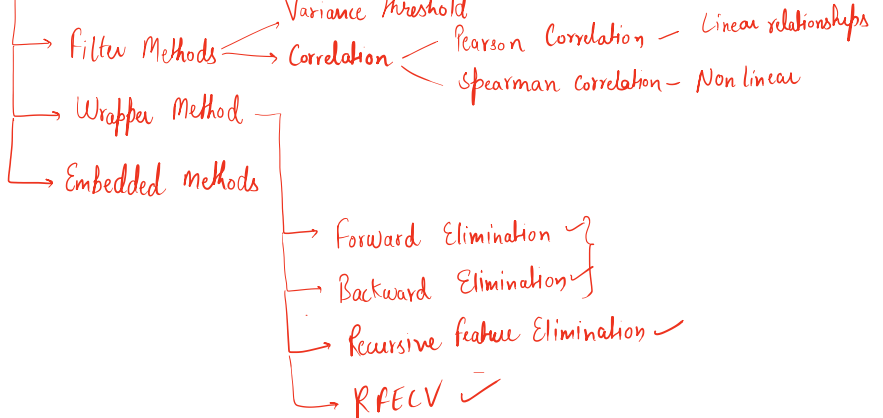
◆ TAKEAWAY

Feature selection tries to remove **redundant** and **irrelevant** inputs without discarding genuinely useful signal.

2.3 Feature selection is not feature transformation

	Feature transformation	Feature selection
Core action	Create a new representation, such as x^2 from x .	Retain only some existing candidate columns.
Effect on candidate space	Expands or changes the representation.	Reduces the representation.
Can both appear in one pipeline?	Yes. Create meaningful candidates first, then select among them.	

Feature Selection



12P-1

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SECTION 3 Why Feature Selection Can Improve Learning

3.1 Generalization and the bias-variance connection

More features increase a model's flexibility. While useful features reduce bias, irrelevant features cause the model to fit random noise, increasing _____ and hurting validation performance.

Classroom question: Training error decreases after ten features are added, but validation error increases. Did the new features help the model learn the stable pattern?

3.2 The curse of dimensionality

Imagine dividing the range of every feature into b equal bins. With p features, the number of possible grid cells is:

Features p	Bins per feature b	Possible cells b^p
2	10	100
4	10	10,000
6	10	1,000,000

$10^2 = 100$
cells
 $p = 2$

$10^4 = 10,000$ cells
 $p = 4$

$10^6 = 1,000,000$ cells
for only 1,000 observations
 $p = 6$

With 1,000 observations and six features, most of the one million cells are empty. Data coverage becomes sparse, so stable patterns are harder to estimate.

3.3 Computation, Memory, and Interpretability

- **Efficiency:** Smaller matrices, faster training/prediction, lower memory use.
- **Explainability:** A model with fewer features is easier to explain.

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$$\frac{p(p+1)}{2} \rightarrow \text{from sklearn. Preprocessing}$$

RFE $\rightarrow$ Initially = 10 Keep = 5

Step 1 $\rightarrow$ Train model on K features

Step 2 $\rightarrow$ Compute R^2 Score

Step 3 $\rightarrow$ Drop one feature with lowest feature importance

Step 4 $\rightarrow$ Will repeat till feature I need

$10 \rightarrow$ Model train - 10 features $\rightarrow$

$\rightarrow 9$
 $\rightarrow 8$
 $\rightarrow 7$
 $\rightarrow 6$
 $\rightarrow 5$

Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{10} x_{10}$$

model. coeff $\rightarrow \beta_1, \beta_2, \dots, \beta_{10}$

- **Stability:** Removing redundant inputs reduces multicollinearity, making linear coefficients more stable.
- Multicollinearity may be less damaging for pure prediction than for coefficient interpretation, but it still represents redundancy and instability.

SECTION 4 The Two Feature-Selection Families

Family	Core question	How selection happens	Cost
Filter	Does each feature pass a statistical or structural screen?	Score or inspect features before fitting the final model.	Low
Wrapper	Which subset gives the chosen model the best validation performance?	Train and compare the model on candidate subsets.	High

Visual: Screening first versus model-in-the-loop search

FILTER All features $\rightarrow$ Simple screen $\rightarrow$ Filtered subset

WRAPPER Candidate subset $\rightarrow$ Train model $\rightarrow$ Validation score

next subset

PRACTICE P1: FEATURE TRIAGE AND FOUNDATIONS

(a) For each feature, decide whether a simple filter should **Remove**, **Flag for investigation**, or **Keep** for model-based evaluation. Give one reason.

Feature	Observed property	Decision	Reason
CGPA_copy	Exactly equals CGPA in every row.	Remove	same info
Student_ID	Every value is unique; variance is high.	remove	not able to identify pattern/no corr with target
Scholarship_flag	Only 2% of students have value 1.	Info	
Random_number	High variance and near-zero target correlation.		
Project_depth ²	Highly correlated with project depth.		

(b) Choose all correct options.

- (i) A feature's usefulness can change when the target or model changes. ☐
- (ii) A high-variance column must be predictive. ☐
- (iii) Feature transformation and feature selection can both appear in one pipeline. ☐
- (iv) The smallest possible subset is always the best subset. ☐
- (v) If two subsets have nearly equal cross-validated performance, the smaller subset is usually preferred. ☐

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(c) With $b = 10$ bins per feature, complete the pattern:

$$p = 3: 10^3 = \quad \quad \quad p = 5: 10^5 = \quad \quad \quad$$

$$p = 7: 10^7 = \quad \quad \quad$$

What happens to data coverage as p grows while the number of observations stays fixed?

(d) A model's training error falls after ten new columns are added, but validation error rises. Diagnose the change using the bias-variance language from Lecture 9.

PART II: FILTER METHODS – SCREEN FEATURES BEFORE THE FINAL MODEL

SECTION 5 Filter Intuition and Duplicate Removal

▷ HOOK

A filter method acts like an **admission gate**. It evaluates a feature using a simple structural or statistical rule and decides whether that feature deserves to enter the modeling stage. The final predictive model is not repeatedly trained during this screening step.

Some filters do not use the target; others compare a feature with the target.

Filter type	Uses target y ?	Examples in this lecture
Structural / unsupervised screen	No	Exact duplicate removal; variance threshold
Target-aware univariate screen	Yes	Pearson correlation between one feature and y
Pairwise input screen	No target required	Correlation between input features

5.1 Removing exact duplicate features

If two columns contain exactly the same value in every row, they provide the same information:

$$f_A = f_B \text{ for every observation.}$$

Keeping both does not create a second independent source of evidence. In the stipend example, CGPA and CGPA_copy are exact duplicates. For a linear model, retaining both also creates _____ in their individual contributions cannot be uniquely separated.

Which copy should be kept?

- Keep the column with the clearer and more meaningful name.
- Prefer the column with fewer missing or corrupted values.
- Prefer the feature that is easier and cheaper to collect at prediction time.

- Check that the apparent duplicate is not a data-leakage column created after the target became known.

◆ TAKEAWAY

Duplicate removal is safe because it removes **repeated information**, not unique information.

SECTION 6 Variance Threshold

▷ HOOK

A feature that barely changes across observations cannot distinguish one observation from another. The simplest case is a constant column: if every student has the same value, that feature cannot explain why stipends differ.

For feature f_j , a variance measure is

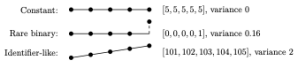
$$\text{Var}(f_j) = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

Choose a threshold τ . Remove the feature when

$$\text{Var}(f_j) < \tau$$

The threshold is a _____, not a universal constant.

Visual: What a variance filter sees



Feature values	Variance	What the filter sees	What must be remembered
[5, 5, 5, 5, 5]	0	No variation; remove.	A constant feature cannot separate observations.
[0, 0, 0, 0, 1]	0.16	Low variation; may be removed for a high τ .	A rare event can still be strongly predictive.
[101, 102, 103, 104, 105]	2	Moderate variation; may be kept.	An identifier can vary greatly and still be irrelevant.

⚠ WARNING – SCALING CLARIFICATION

Zero-variance detection does not require scaling. For a nonzero threshold, raw variances are affected by units, so features must be placed on a comparable range or given domain-specific thresholds. Do **not** standardize every continuous feature to variance 1 and then apply a variance threshold; that would erase the differences the method is trying to inspect.

For a binary feature taking values 0 and 1, if the proportion of ones is q , then

$$\text{Var}(\text{binary feature}) = q(1 - q)$$

A rare binary feature has low variance because q is near 0 or 1. Low variance alone is therefore not enough to call it useless.

Limitations of a variance threshold

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$$\frac{1}{n} \sum (x_{ij} - \bar{x}_j)^2$$

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- A high-variance feature may have no relationship with the target.
- A low-variance feature may be highly predictive, especially for a rare but important condition.
- The **method** ignores combinations and interactions among features.
- The threshold depends on units, scaling choices, and context.

SECTION 7 Pearson Correlation Between Each Feature and the Target

▷ HOOK

Variance tells us whether a feature changes. It does not tell us whether the feature changes **together with the target**. For a numerical feature and numerical target, Pearson correlation measures the strength and direction of a **linear** relationship.

$$r_{xy} = \frac{\sum_i (x_{ij} - \bar{x}_j)(y_i - \bar{y})}{\sqrt{\sum_i (x_{ij} - \bar{x}_j)^2 \sum_i (y_i - \bar{y})^2}}$$

- r close to +1: strong positive linear relationship.
- r close to -1: strong negative linear relationship.
- r close to 0: weak linear relationship, **not necessarily no relationship**.

An illustrative rule is

$$\text{Remove } f_j \text{ when } |r_{xy}| < 0.30$$

The value 0.30 is only an illustration; the threshold must be validated through model performance.

Feature	Illustrative r with stipend	Decision under $ r \geq 0.30$	Caution
CGPA	0.68	✓	Strong linear association in this sample.
IQ score	0.18	✗	Could still help jointly or nonlinearly.
Number of projects	0.57	✓	Useful linear association.
Attendance	0.09	✗	Threshold decision must be validated.
Student_ID	-0.02	✗	Large variance did not imply relevance.

7.1 The nonlinear counterexample

Let $y = x^2$ for symmetric values of x :

$$\begin{array}{c|cccc} x & -2 & -1 & 0 & 1 & 2 \\ \hline y = x^2 & 4 & 1 & 0 & 1 & 4 \end{array}$$

- Calculate $\bar{x}$ and $\bar{y}$:
 $\bar{x} = 0$, $\bar{y} = 2$.
- Compute the numerator $\sum_i (x_i - \bar{x})(y_i - \bar{y})$:
 $\sum_i (x_i - 0)(y_i - 2) = (-2)(-2) + (-1)(-1) + (0)(-2) + (1)(-1) + (2)(2)$
 $= -4 + 1 + 0 - 1 + 4$
 $= 0$.
- Therefore, $r_{xy} = 0$.
A $|r| \geq 0.30$ filter would _____, even though y is completely determined by x .

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$$x_1 \text{ and } x_2$$

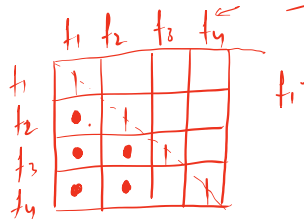
$$10.89$$

$$\left. \begin{array}{l} -0.89 \\ 0.89 \end{array} \right\}$$

Target - Stipend

heatmap → df.corr()

↑ Pearson



★ KEY INSIGHT

A feature can have zero Pearson correlation and still be perfectly predictive through a nonlinear relationship. Pearson correlation measures **linear association only**.

Limitations of target correlation

- It can miss nonlinear relationships.
- It evaluates one feature at a time and can miss useful interactions.
- It is sensitive to outliers.
- Its threshold is context-dependent.
- Correlation does not establish causation or future stability.

SECTION 8 Correlation Among Input Features

The previous calculation compared each feature with y . Correlation can also compare input features with one another to detect redundancy and reduce multicollinearity.

Suppose f_1 and f_2 have correlation 0.95. They carry very similar linear information. A practical rule is to keep one representative feature, but the choice should be deliberate.

How to choose which correlated feature to keep

- Prefer the feature with the stronger validated relationship with the target.
- Prefer the feature with less missingness and measurement noise.
- Prefer the feature that is easier to interpret and collect.
- Use domain knowledge before removing intentionally engineered terms.

⚠ WARNING – DO NOT APPLY PAIRWISE RULES BLINDLY

...

If the features x and x' can be correlated, our data may be needed to represent curvature. A high input correlation does not automatically prove that one engineered feature should be removed. Check the final decision through validation performance.

8.1 Filter-method summary

Strengths	Limitations
Simple, fast, and scalable.	Usually ignores interactions among multiple features.
Can remove obvious problems before expensive modeling.	A statistical score may not match the needs of the final model.
Often model-independent and easy to explain.	Threshold choice can be arbitrary or unstable.
Useful as the first stage of a larger pipeline.	Univariate linear measures can miss nonlinear signal.

◆ TAKEAWAY

Filter methods are efficient, but they do not directly ask whether the chosen model performs better with a particular combination of features.

PRACTICE P2 - FILTER-METHOD DETECTIVE

- (a) Use $x = [-2, -1, 0, 1, 2]$ and $y = x^2 = [4, 1, 0, 1, 4]$.
 (i) $\bar{x} =$ _____ and $\bar{y} =$ _____.
 (ii) The covariance numerator is _____, so Pearson correlation is _____.

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- (iii) A $|r| \geq 0.30$ rule would _____.
 (iv) Explain why that decision is wrong for prediction.

(b) Match the filter to the issue.

Observed issue	Most direct screen
1 Two columns are equal in every row.	A Target correlation
2 A column is constant.	B Input-input correlation
3 A numeric feature has weak linear association with numeric y .	C Duplicate removal
4 Two predictors have correlation 0.97.	D Variance threshold

Answers: 1 $\rightarrow$ _____, 2 $\rightarrow$ _____, 3 $\rightarrow$ _____, 4 $\rightarrow$ _____.

- (c) Find the mistake. "Student_ID has very high variance, so a variance filter proves that it is useful for predicting stipend."

 (d) Scholarship_flag equals 1 for only 3% of students. Should it be removed solely because its variance is low? [] Justify.

 (e) Project_depth and Project_depth² are highly correlated. A student immediately removes the squared term. Is this decision justified? [] Justify.

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PART III: WRAPPER METHODS - LET MODEL PERFORMANCE JUDGE THE SUBSET

SECTION 9 The Wrapper Idea and the Wrapper Loop

◆ HOOK

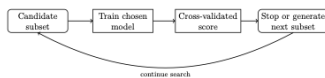
A feature can look weak by itself and become useful when paired with another feature. Conversely, two individually useful features may become redundant when used together. Wrapper methods evaluate whole subsets with the predictive model we actually plan to use.

★ KEY INSIGHT

A wrapper method "wraps" model training and validation around the feature-selection process.

9.1 The wrapper loop

1. Generate a candidate subset of features.
2. Train the chosen model on that subset.
3. Evaluate the subset using validation data or cross-validation.
4. Generate the next candidate subset according to the search strategy.
5. Stop when the search is complete or a stopping rule is met.



⚠ WARNING - DO NOT SELECT WITH RAW TRAINING R^2

For ordinary least squares on the same training data, adding predictors cannot make training R^2 worse. A wrapper should use cross-validated R^2 , cross-validated RMSE, or another validation metric aligned with the problem. Adjusted R^2 can be a secondary linear-regression diagnostic, but cross-validation is the general rule used here.

SECTION 10 Exhaustive Feature Selection

Exhaustive search evaluates every nonempty subset and chooses the subset with the best validation score. Each of p features has two states: included or excluded. Therefore,

$$\text{Number of nonempty subsets} = 2^p - 1$$

Candidate features p	Nonempty subsets $2^p - 1$
4	15
20	1,048,575
30	1,073,741,823

Why exhaustive search becomes impractical

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f_1, f_2, f_3, f_4
 $\uparrow$
 $\{f_1, f_2\}$
 $\uparrow$
 $\{f_1, f_3\}$
 $\uparrow$
 $\{f_1, f_4\}$
 $\uparrow$
 $2^2 - 1$
 $\{ \}$
 $\uparrow$
 $p = f_1, f_2$
 $\uparrow$
 $\{f_1\}, \{f_2\}, \{f_1, f_2\}$

- **Advantage:** if every subset is evaluated correctly, the best subset in the search space can be found.
- **Disadvantage:** computational cost grows exponentially.
- **Disadvantage:** testing many subsets can overfit the validation procedure and select a chance winner.
- **Disadvantage:** the result depends on the chosen model and evaluation metric.

SECTION 11 Backward Elimination

HOOK

Backward elimination starts with all features. At each stage, remove each remaining feature one at a time, evaluate the resulting subsets, and keep the best candidate for the next stage.

Treat every score below as cross-validated R^2 .

11.1 Start with all four features

Current subset	Cross-validated R^2
$\{f_1, f_2, f_3, f_4\}$	0.89

11.2 Remove one feature from the four-feature set

Candidate subset	Cross-validated R^2
$\{f_2, f_3, f_4\}$	0.81
$\{f_1, f_3, f_4\}$	0.71
$\{f_1, f_2, f_4\}$	0.91
$\{f_1, f_2, f_3\}$	0.65

Continue with $\{f_1, f_2, f_4\}$; the removed feature is f_3 . Notice that validation performance improved from 0.89 to 0.91.

11.3 Repeat on $\{f_1, f_2, f_4\}$

Candidate subset	Cross-validated R^2
$\{f_2, f_4\}$	0.79
$\{f_1, f_4\}$	0.81
$\{f_1, f_2\}$	0.83

Continue with $\{f_1, f_2\}$; remove f_4 .

11.4 Compare the remaining single-feature models

Candidate subset	Cross-validated R^2
$\{f_2\}$	0.63
$\{f_1\}$	0.53

The path therefore keeps $\{f_2\}$ at the one-feature stage.

11.5 Choose the best subset encountered along the path

Best subset at that size	Cross-validated R^2
$\{f_1, f_2, f_3, f_4\}$	0.89
$\{f_1, f_2, f_4\}$	0.91
$\{f_1, f_2\}$	0.83
$\{f_2\}$	0.63

The selected subset is $\{f_1, f_2, f_4\}$ because it achieved the highest cross-validated R^2 ; it is not selected merely because it is the final or smallest subset.

11.6 Computational advantage and greedy limitation

For $p = 4$, backward elimination fits

$$1 + 4 + 3 + 2 = 10$$

candidate models instead of all 15 nonempty subsets. In general,

$$1 + p + (p-1) + \dots + 2 = \frac{p(p+1)}{2}$$

This grows quadratically rather than exponentially, although every score may still require several cross-validation fits.

WARNING — GREEDY SEARCH

Backward elimination makes a locally best decision at each stage. Once a feature is removed, it is never reconsidered. The method can therefore miss a globally best combination that contains an earlier discarded feature.

SECTION 12 Sequential Forward Selection

HOOK

Forward selection begins with no input features. For regression, the baseline model predicts the mean target $\bar{y}$. The method adds the feature that gives the largest validation improvement, fixes that choice, and repeats.

12.1 Baseline model

$$\text{No input feature} \Rightarrow \hat{y} = \bar{y}$$

12.2 Add one feature

Candidate subset	Cross-validated R^2
$\{f_1\}$	0.63
$\{f_2\}$	0.51
$\{f_3\}$	0.43
$\{f_4\}$	0.49

Select f_1 .

12.3 Add one feature to $\{f_1\}$

Candidate subset	Cross-validated R^2
$\{f_1, f_2\}$	0.63
$\{f_1, f_3\}$	0.71
$\{f_1, f_4\}$	0.80

Select f_1, f_4 .

12.4 Add one feature to $\{f_1, f_4\}$

Candidate subset	Cross-validated R^2
$\{f_1, f_4, f_2\}$	0.81
$\{f_1, f_4, f_3\}$	0.85

Select f_1, f_4, f_3 .

12.5 Try the final feature

Candidate subset	Cross-validated R^2
$\{f_1, f_2, f_3, f_4\}$	0.83

The all-feature model scores 0.83, below the best three-feature score 0.85. The best subset encountered is therefore $\{f_1, f_4, f_3\}$.

TAKEAWAY

Forward selection does not automatically keep adding features until all are used. Stop when an additional feature no longer gives a meaningful validation improvement, or choose the best subset along the entire forward path.

12.6 When forward or backward selection is more natural

- Use forward selection when p is large and the desired final subset is small; for example, keep about 10 of 100 features.

- Use _____, when the model can fit all p features and the desired final subset retains most of them; for example, keep about 90 of 100 features.
- Both are greedy and model-specific, so they can select different subsets.

12.7 Wrapper-method summary

Strengths	Limitations
Directly optimizes performance for the chosen model and metric.	Computationally expensive, especially with cross-validation.
Can capture useful combinations and interactions that filters miss.	Can overfit the feature-search procedure.
Produces a subset tailored to the final modeling goal.	Selected features may not transfer to a different model.
Sequential methods are much cheaper than exhaustive search.	Greedy methods can miss the globally best subset.

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PRACTICE P3 WRAPPER SEARCH AND REASONING

(a) The current backward-elimination subset is $\{f_1, f_2, f_4\}$. Removing one feature gives:

Candidate subset	Cross-validated R^2
$\{f_2, f_4\}$	0.79
$\{f_1, f_4\}$	0.81
$\{f_1, f_2\}$	0.83

- Which subset continues? _____
- Which feature is removed? _____
- Why can the procedure still miss the globally best subset? _____

(b) For $p = 6$, compare the number of candidate models:

Exhaustive: $2^6 - 1 = \underline{\hspace{2cm}}$. Backward path: $\frac{6(6+1)}{2} = \underline{\hspace{2cm}}$.

State the growth pattern of each method.

(c) Arrange the wrapper loop in the correct order. Write the letters in sequence.

- Evaluate with validation data or cross-validation.
- Generate the next candidate subset.
- Train the chosen model on the candidate subset.
- Stop when the search or stopping rule is complete.
- Generate an initial candidate subset.

Correct order: _____

(d) **Assertion-Reason.**

A: If exhaustive search evaluates every nonempty subset correctly, it can find the best subset in that search space.

R: Exhaustive search evaluates all $2^p - 1$ nonempty subsets.

- Both A and R are true, and R is the correct explanation of A. []
- Both A and R are true, but R is not the correct explanation of A. []
- A is true but R is false. []
- A is false but R is true. []

(e) Match the search direction to the goal.

Goal	More natural method
From 100 features, keep about 10.	_____
From 100 features, retain about 90.	_____

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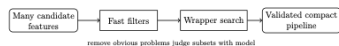
PART IV: CHOOSING A METHOD AND VALIDATING THE DECISION

SECTION 13 Filter and Wrapper Methods Compared Directly

Criterion	Filter	Wrapper
Uses final model performance directly?	Usually no	Yes
Captures feature interactions?	Usually no	Yes, if the chosen model can represent them
Model-specific?	Usually no	Yes
Computational cost	Low	High
Best role	Fast screening and preprocessing	Careful subset search for a chosen model

13.1 Practical method-choice rules

- Start with filters when p is very large or when obvious duplicates and near-constant columns exist.
- Use a wrapper when the feature count is manageable, model-specific performance matters, and computational budget is available.
- Combine methods when useful: apply simple filters first, then run a wrapper on the reduced set.
- Always compare the selected-feature pipeline with an all-feature baseline using the same validation procedure.



SECTION 14 Validation Discipline: Avoid the Leakage Trap

► HOOK

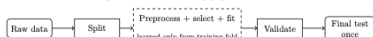
Feature selection is part of model training. It is not a harmless preliminary look at the whole dataset.

⚠ WARNING – THE LEAKAGE TRAP

Wrong: compute correlations or choose the number of features using the entire dataset, and then report cross-validation on those already-selected columns. The validation folds have influenced selection.

Right: inside every fold, learn preprocessing and selection rules using only that fold's training portion, then apply the fitted rules to the corresponding validation portion.

Visual: Selection belongs inside the training pipeline



The test set is not used to choose thresholds, feature counts, the model, or stopping rules. It is used once, after all decisions are fixed, to estimate final generalization.

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SECTION 15 The Selected Subset Is Not Universal

A selected feature set is supported by a particular experimental design; it is not a permanent truth.

- Change the **target**, and relevant features can change.
 - Change the **model**, and the useful subset can change.
 - Change the **evaluation metric**, and the preferred subset can change.
 - Change the **sample**, and unstable selection procedures may choose different features.
- Therefore, report the selected features together with the dataset, target, model, metric, and validation design that supported them.

Fill the Feedback

SECTION 16 End-to-End Feature-Selection Workflow

- Step 1:** Define the target, evaluation metric, and final modeling goal.
- Step 2:** Create the train-validation-test structure before learning selection rules.
- Step 3:** Remove exact duplicates and structurally unusable columns.
- Step 4:** Use simple filters for obvious low-information or redundant inputs, while checking their limitations.
- Step 5:** Use a wrapper when model-specific subset evaluation is needed and computationally feasible.
- Step 6:** Choose thresholds, feature counts, and stopping rules inside cross-validation.
- Step 7:** Compare with the all-feature baseline; prefer the simpler model only when generalization is maintained or improved.
- Step 8:** Evaluate once on the untouched test set and report the final feature set with the method used to obtain it.

◆ TAKEAWAY

Feature selection is successful only when the smaller representation preserves useful signal and improves the reliability, efficiency, or interpretability of the complete predictive pipeline.

16.1 Closing synthesis

1. More features can add signal, but they can also add redundancy, noise, computation, and variance.
2. Filter methods screen quickly using structural or statistical rules, but usually miss interactions.
3. Wrapper methods judge candidate subsets using the chosen model, but repeated fitting is expensive and greedy searches can miss the global optimum.
4. Selection must occur inside validation; the test set remains untouched until every decision is fixed.

PRACTICE P4 COMPREHENSIVE CHALLENGE

- (a) **Diagnose the pipeline.** A student performs these steps:
- (1) Compute correlations using the entire dataset.
 - (2) Keep the top five features.
 - (3) Run five-fold cross-validation using those five columns.
 - (4) Report the mean validation score.
- Is the reported score trustworthy? [] Explain the problem and correct the procedure.
- _____
- _____
- _____

- (b) Choose the most appropriate starting strategy for each situation.

Situation	Starting strategy
Thousands of columns contain exact duplicates and near-constant fields.	_____
A manageable feature set may contain model-specific interactions, and sufficient computation is available.	_____
Obvious problems can be removed cheaply, after which a smaller model-aware search is feasible.	_____

- (c) **True or False.** Justify each answer briefly.

Statement	T/F	Justification
The smallest possible feature set is always best.	[]	_____
A low-variance feature is always useless.	[]	_____
Wrapper methods usually cost more because they repeatedly fit and evaluate models.	[]	_____
Two different models can legitimately select different features from the same dataset.	[]	_____
The test set may be used to choose the correlation threshold.	[]	_____

- (d) Arrange the following stages into a trustworthy sequence.
- A. Evaluate once on the untouched test set.
 - B. Split the data before learning selection rules.
 - C. Define the target, metric, and modeling goal.
 - D. Fit preprocessing, selection, and the model inside cross-validation.
 - E. Compare with the all-feature baseline and fix all choices.
- Correct order: _____.
- (e) Two subsets have nearly the same cross-validated performance. One uses 18 features and the other uses 7. Which would you normally prefer, and what could justify the larger subset?
- _____
- _____
- (f) Why can the selected features change when the target, model, metric, or sample changes?
- _____
- _____

BY THE END OF THIS SHEET, YOU CAN

- ☐ Explain feature selection as choosing a subset of existing inputs.
- ☐ Distinguish relevant, redundant, and irrelevant features.
- ☐ Connect unnecessary features with variance, sparse high-dimensional coverage, computation, and interpretability.
- ☐ Distinguish feature selection from feature transformation.
- ☐ Apply duplicate removal, variance threshold, target correlation, and input-correlation screens with appropriate cautions.
- ☐ Explain why zero Pearson correlation does not imply zero predictive value.
- ☐ Calculate the number of exhaustive subsets and trace backward and forward sequential searches.
- ☐ Compare filter and wrapper methods by cost, interaction-awareness, and model dependence.
- ☐ Place selection correctly inside cross-validation and protect the final test set.
- ☐ Report a selected subset as conditional on the target, model, metric, sample, and validation design.